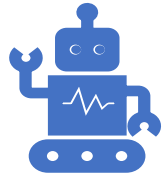


Section 6

Systems Programming ECE3502

Assembler Features

- These are the features which **do not depend on the architecture of the machine**
- Such features are more related to software than to machine architecture



Machine Dependent Assembler Features

- 1) Instruction formats and addressing modes (SIC/XE)
- 2) Program relocation



Machine Independent Assembler Features

- 1) Literals
- 2) Symbol-defining statements
- 3) Expressions
- 4) Program blocks
- 5) Control sections and program linking

Literals

5	COPY	START	0
10	FIRST	STL	RETADR
12		LDB	#LENGTH
13		BASE	LENGTH
15	CLOOP	+JSUB	RDREC
20		LDA	LENGTH
25		COMP	#0
30		JEQ	ENDFIL
35		+JSUB	WRREC
40		J	CLOOP
45	ENDFIL	LDA	EOF
50		STA	BUFFER
55		LDA	#3
60		STA	LENGTH
65		+JSUB	WRREC
70		J	@RETADR
80	EOF	BYTE	C'EOF'
95	RETADR	RESW	1
100	LENGTH	RESW	1
105	BUFFER	RESB	4096

5	COPY	START	0
10	FIRST	STL	RETADR
13		LDB	#LENGTH
14		BASE	LENGTH
15	CLOOP	+JSUB	RDREC
20		LDA	LENGTH
25		COMP	#0
30		JEQ	ENDFIL
35		+JSUB	WRREC
40		J	CLOOP
45	ENDFIL	LDA	=C'EOF'
50		STA	BUFFER
55		LDA	#3
60		STA	LENGTH
65		+JSUB	WRREC
70		J	@RETADR
93		LTORG	
95	RETADR	RESW	1
100	LENGTH	RESW	1
105	BUFFER	RESB	4096
106	BUFEND	EQJ	*
107	MAXLEN	EQJ	BUFEND-BUFFER ⁴⁷

Literals

- It is easy for a programmer to write **the value of a constant operand** as part of the instruction that uses it.
- This **avoids defining the constant elsewhere in the program** and making a label for it. Such an operand is called **a literal** because the value is stated literally in the instruction.
- A literal is defined with **a prefix = followed by a specification of the literal value**.

Example:

001A	ENDFIL	LDA	=C'EOF'	032010
.				
002D	*	=	C'EOF'	454F46

Difference between Literal and Immediate operand

- In case of **literals** the assembler generates **the specified value as a constant at some other memory location**.
- In the Immediate mode **the operand value is assembled as part of the instruction itself**.

Example:

0010		LDA	#03	010003
....				
001A	ENDFIL	LDA	=C'EOF'	032010
....				
002D	*		= C'EOF'	454F46

Literal Pool

- All the literal operands used in a program are gathered together into one or more **literal pools**.
- This is usually placed **at the end of the program**.
- Ex: 1076 * =X'05'
- In some cases, it is **placed at some other location in the object program**. An assembler directive **LTORG** is used
- Whenever **LTORG** is encountered, it creates a literal pool that contains all the literal operands used since the beginning of the program.
- The literal pool definition is done after **LTORG** is encountered. It is better to place the literals close to the instruction

Literal Pool

001A	ENDFIL	LDA	=C'EOF'
.			
.			
.			
002A		J	RETADR
		LTORG	
002D	*	=C'EOF'	
.			
.			
.			
		LDA	=C'EOF'

Opcode 6 bits						n	i	x	b	p	e	displacement
0	0	0	0	0	0	1	1	0	0	1	0	(010)

2D-1D

We don't need to
generate another
literal

Recognizing Duplicate Literals

- Duplicate literals can be identified by **comparing character strings** (If the literal text is exactly the same, the assembler recognizes it immediately.) Ex: X'05'.

200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER			
205	.				
210	105D	WRREC	CLEAR	X	B410
212	105F		LOT	LENGTH	774000
215	1062	WLOOP	TD	=X'05'	E32011
220	1065		JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	DF2008
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSJB		4F0000
255			END	FIRST	
1076	*		=X'05'		05

- Sometimes literals look different but represent **the same value**, thus the generated value can be compared.
Ex: The literals =C'EOF' and =X'454F46' are identical operand values.

Literals denoting the value of the location counter

- Literals referring to the current value of the location counter (denoted by the symbol *). Such literals are useful for loading base registers.

Ex:101A	BASE	*	BASE = 101A
101D	LDA	*	Load A with the value stored at address 101D
1020	LDB	=*	load B with a literal equal to the current address "1020"
	:		
	LTORG		
-----	*	=1020	

- Such literal operands will have different values in different places of the program since they hold the current value of the location counter.

Literal Table

- A **literal table** is created for the literals which are used in the program.
- The **literal table** contains the **literal name**, **operand value**, **length** and the **address** assigned to the operand.

```
200      .
205      .
210 105D  WRREC  CLEAR  X      B4110
212 105F      LOT    LENGTH  774000
215 1062  WLOOP  TD      =X'05' E32011
220 1065      JEQ    WLOOP  332FFA
225 1068      LDCH   BUFFER,X 53C003
230 106B  WD      =X'05'  DF2008
235 106E      TIXR   T      B850
240 1070      JLT    WLOOP  3B2FEF
245 1073      RSJB           4F0000
255      END      FIRST
1076 *      =X'05'      05
```

LITTAB

Literal name	Value	Length	Address
=X'05'	05	1	1076

TD =X'05'

Opcode (6)	n	i	x	b	p	e	Disp(12)
E0	1	1	0	0	1	0	1076- 1065=11
1110 00	11		0010				0000 0001 0001
E32011							

WD =X'05'

Opcode (6)	n	i	x	b	p	e	Disp(12)
DC	1	1	0	0	1	0	1076- 106E=8
1101 11	11		0010				0000 0000 1000
DF2008							

Symbol-Defining Statements (EQU/ORG)

EQU:

Symbol EQU value

- This statement defines the given symbol (i.e. enters it into the Symbol Table) and assigning to it the value specified.
- The value can be a **constant or an expression involving constants**.
- Can be used to improve **readability** in place of numeric values

Instead of:

```
+LDT      #4096
```

We can write:

```
MAXLEN    EQU    4096    =    MAXLEN    WORD    4096
+LDT      # MAXLEN
```

Symbol-Defining Statements (EQU/ORG)

One **restriction** with the usage of EQU is **whatever symbol occurs in the right-hand side of EQU should be predefined**.

Ex: the following statement is not valid

```
BETA    EQU    ALPHA
ALPHA    RESW    1
```

As the symbol ALPHA is assigned to BETA before it is defined. The value of ALPHA is not known.

CORRECT:

```
1000    ALPHA    RESW    1
```

```
BETA    EQU    ALPHA
```

BETA=1000

Symbol-Defining Statements (EQU/ORG)

ORG:

- Can be used to indirectly assign values to the symbol (changes the value in the location counter).
- ORG can be used in label definition.
- While using ORG, the symbol occurring in the statement should be predefined as is required in EQU statement.

```
EX:          ORG      ALPHA
             BYTE1    RESB    1
             BYTE2    RESB    1
             ORG
             ALPHA    RESB    1
```

NOT ALLOWED, forward referencing

Expressions

- Can be used in the place of operands in the instruction, it should be evaluated to generate a value or address.
- Arithmetic operators: +, -, *, /

Expressions:

→ **Absolute:** Constant Value or relative terms

MAXLEN EQU 1000

MAXLEN EQU BUFEND – BUFFER (NO mul or div in relative terms)

→ **Relative:** Depends on the starting address of the program

MAXLEN EQU LENGTH+(BUFEND – BUFFER) (NO mul or div in relative terms)

→ **Neither Absolute Nor Relative:** Consider as errors

BUFEND + BUFFER

100 – BUFFER

3* BUFFER

Symbol	Type	Value
RETADR	R	0030
BUFFER	R	0036
BUFEND	R	1036
MAXLEN	A	1000

Pass One

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	??	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14			BASE LENGTH	
15		CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45		ENDFIL	LDA =C' EOF'	
50			STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93			LTORG	
		*	=C' EOF'	
95	0030	RETADR	RESW 1	
100	0033	LENGTH	RESW 1	
105	0036	BUFFER	RESB 4096	
106		BUFEND	EQU *	
107		MAXLEN	EQU BUFEND-BUFFER	
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125		RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210		WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215		WLOOP	TD =X'05'	
220			JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255			END FIRST	
		*	=X'05'	

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	??		BASE	LENGTH	
15		CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45		ENDFIL	LDA	=C' EOF'	
50			STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93			LTORG		
		*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106		BUFEND	EQU	*	
107		MAXLEN	EQU	BUFEND-BUFFER	
110		.			
115		.	SUBROUTINE TO READ RECORD INTO BUFFER		
120		.			
125		RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195		.			
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205		.			
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	??	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45		ENDFIL	LDA	=C' EOF'	
50			STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93			LTORG		
		*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106		BUFEND	EQU	*	
107		MAXLEN	EQU	BUFEND-BUFFER	
110		.			
115		.	SUBROUTINE TO READ RECORD INTO BUFFER		
120		.			
125		RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A, S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER, X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195		.			
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205		.			
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER, X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	??	ENDFIL	LDA	=C' EOF'	
50			STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93			LTORG		
		*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106		BUFEND	EQU	*	
107		MAXLEN	EQU	BUFEND-BUFFER	
110		.			
115		.	SUBROUTINE TO READ RECORD INTO BUFFER		
120		.			
125		RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A, S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER, X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195		.			
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205		.			
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER, X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	??		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93			LTORG		
		*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106		BUFEND	EQU	*	
107		MAXLEN	EQU	BUFEND-BUFFER	
110		.			
115		.	SUBROUTINE TO READ RECORD INTO BUFFER		
120		.			
125		RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195		.			
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205		.			
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C' EOF'	
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	??		LTORG	
		*	=C' EOF'	
95	0030	RETADR	RESW 1	
100	0033	LENGTH	RESW 1	
105	0036	BUFFER	RESB 4096	
106		BUFEND	EQU *	
107		MAXLEN	EQU BUFEND-BUFFER	
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125		RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210		WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215		WLOOP	TD =X'05'	
220			JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255			END FIRST	
		*	=X'05'	

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C'EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	??	*	=C'EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106		BUFEND	EQU	*	
107		MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125		RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	??	BUFEND	EQU	*	
107		MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125		RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	1036	BUFEND	EQU	*	
107	??	MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125		RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	1036	BUFEND	EQU	*	
107	1000	MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125	??	RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210		WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215			TD	=X'05'	
220		WLOOP	JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JBQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	1036	BUFEND	EQU	*	
107	1000	MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125	1036	RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JBQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JBQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210	??	WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215		WLOOP	TD	=X'05'	
220			JBQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	1036	BUFEND	EQU	*	
107	1000	MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125	1036	RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210	105D	WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215	??	WLOOP	TD	=X'05'	
220			JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	1036	BUFEND	EQU	*	
107	1000	MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125	1036	RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210	105D	WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215	1062	WLOOP	TD	=X'05'	
220	??		JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255			END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	1036	BUFEND	EQU	*	
107	1000	MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125	1036	RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210	105D	WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215	1062	WLOOP	TD	=X'05'	
220	1065		JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255	??		END	FIRST	
		*	=X'05'		

Pass One

Line	Loc	Source statement			Object code
5	0000	COPY	START	0	
10	0000	FIRST	STL	RETADR	17202D
13	0003		LDB	#LENGTH	69202D
14	-----		BASE	LENGTH	
15	0006	CLOOP	+JSUB	RDREC	4B101036
20	000A		LDA	LENGTH	032026
25	000D		COMP	#0	290000
30	0010		JEQ	ENDFIL	332007
35	0013		+JSUB	WRREC	4B10105D
40	0017		J	CLOOP	3F2FEC
45	001A	ENDFIL	LDA	=C' EOF'	
50	001D		STA	BUFFER	0F2016
55	0020		LDA	#3	010003
60	0023		STA	LENGTH	0F200D
65	0026		+JSUB	WRREC	4B10105D
70	002A		J	@RETADR	3E2003
93	-----		LTORG		
	002D	*	=C' EOF'		
95	0030	RETADR	RESW	1	
100	0033	LENGTH	RESW	1	
105	0036	BUFFER	RESB	4096	
106	1036	BUFEND	EQU	*	
107	1000	MAXLEN	EQU	BUFEND-BUFFER	
110	.				
115	.		SUBROUTINE TO READ RECORD INTO BUFFER		
120	.				
125	1036	RDREC	CLEAR	X	B410
130	1038		CLEAR	A	B400
132	103A		CLEAR	S	B440
133	103C		+LDT	#MAXLEN	75101000
135	1040	RLOOP	TD	INPUT	E32019
140	1043		JEQ	RLOOP	332FFA
145	1046		RD	INPUT	DB2013
150	1049		COMPR	A,S	A004
155	104B		JEQ	EXIT	332008
160	104E		STCH	BUFFER,X	57C003
165	1051		TIXR	T	B850
170	1053		JLT	RLOOP	3B2FEA
175	1056	EXIT	STX	LENGTH	134000
180	1059		RSUB		4F0000
185	105C	INPUT	BYTE	X'F1'	
195	.				
200	.		SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.				
210	105D	WRREC	CLEAR	X	B410
212	105F		LDT	LENGTH	774000
215	1062	WLOOP	TD	=X'05'	
220	1065		JEQ	WLOOP	332FFA
225	1068		LDCH	BUFFER,X	53C003
230	106B		WD	=X'05'	
235	106E		TIXR	T	B850
240	1070		JLT	WLOOP	3B2FEF
245	1073		RSUB		4F0000
255	-----		END	FIRST	
	??	*	=X'05'		

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW 1	
100	0033	LENGTH	RESW 1	
105	0036	BUFFER	RESB 4096	
106	1036	BUFFEND	EQU *	
107	1000	MAXLEN	EQU BUFEND-BUFFER	
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	??
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW 1	
100	0033	LENGTH	RESW 1	
105	0036	BUFFER	RESB 4096	
106	1036	BUFFEND	EQU *	
107	1000	MAXLEN	EQU BUFEND-BUFFER	
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	-----
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	??
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW 1	
100	0033	LENGTH	RESW 1	
105	0036	BUFFER	RESB 4096	
106	1036	BUFFEND	EQU *	
107	1000	MAXLEN	EQU BUFEND-BUFFER	
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	-----
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	032010
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	??
	002D	*	=C'EOF'	
95	0030	RETADR	RESW 1	
100	0033	LENGTH	RESW 1	
105	0036	BUFFER	RESB 4096	
106	1036	BUFEND	EQU *	
107	1000	MAXLEN	EQU BUFEND-BUFFER	
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	
	1076	*	=X'05'	

Pass Two

Opcode 6 bits	n	i	x	b	p	e	displacement
0000 00	1	1	0	0	1	0	(010)H

Address of literal – PC

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Line	Loc	Source statement		Object code
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
13	0003		LDB	#LENGTH
14	-----		BASE	LENGTH
15	0006	CLOOP	+JSUB	RDREC
20	000A		LDA	LENGTH
25	000D		COMP	#0
30	0010		JEQ	ENDFIL
35	0013		+JSUB	WRREC
40	0017		J	CLOOP
45	001A	ENDFIL	LDA	=C'EOF'
50	001D		STA	BUFFER
55	0020		LDA	#3
60	0023		STA	LENGTH
65	0026		+JSUB	WRREC
70	002A		J	@RETADR
93	-----		LTORG	
	002D	*	=C'EOF'	??
95	0030	RETADR	RESW	1
100	0033	LENGTH	RESW	1
105	0036	BUFFER	RESB	4096
106	1036	BUFFEND	EQU	*
107	1000	MAXLEN	EQU	BUFEND-BUFFER
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR	X
130	1038		CLEAR	A
132	103A		CLEAR	S
133	103C		+LDT	#MAXLEN
135	1040	RLOOP	TD	INPUT
140	1043		JEQ	RLOOP
145	1046		RD	INPUT
150	1049		COMPR	A,S
155	104B		JEQ	EXIT
160	104E		STCH	BUFFER,X
165	1051		TIXR	T
170	1053		JLT	RLOOP
175	1056	EXIT	STX	LENGTH
180	1059		RSUB	
185	105C	INPUT	BYTE	X'F1'
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR	X
212	105F		LDT	LENGTH
215	1062	WLOOP	TD	=X'05'
220	1065		JEQ	WLOOP
225	1068		LDCH	BUFFER,X
230	106B		WD	=X'05'
235	106E		TIXR	T
240	1070		JLT	WLOOP
245	1073		RSUB	
255	-----		END	FIRST
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	-----
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	032010
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	-----
	002D	*	=C'EOF'	454F46
95	0030	RETADR	RESW 1	??
100	0033	LENGTH	RESW 1	
105	0036	BUFFER	RESB 4096	
106	1036	BUFFEND	EQU *	
107	1000	MAXLEN	EQU BUFEND-BUFFER	
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	
	1076	*	=X'05'	

The ASCII values of EOF

Pass Two

Symbol Table			Literal Table		
Symbol	Type	Value	Name	value	Address
FIRST	R	0000	=C'EOF'	454F46	002D
CLOOP	R	0006	=X'05'	05	1076
ENDFIL	R	001A			
RETADR	R	0030			
LENGTH	R	0033			
BUFFER	R	1036			
BUFFEND	R	1036			
MAXLEN	A	1000			
RDREC	R	1036			
RLOOP	R	1040			
EXIT	R	1056			
INPUT	R	105C			
WRREC	R	105D			
WLOOP	R	1062			

Line	Loc	Source statement		Object code
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
13	0003		LDB	#LENGTH
14	-----		BASE	LENGTH
15	0006	CLOOP	+JSUB	RDREC
20	000A		LDA	LENGTH
25	000D		COMP	#0
30	0010		JEQ	ENDFIL
35	0013		+JSUB	WRREC
40	0017		J	CLOOP
45	001A	ENDFIL	LDA	=C'EOF'
50	001D		STA	BUFFER
55	0020		LDA	#3
60	0023		STA	LENGTH
65	0026		+JSUB	WRREC
70	002A		J	@RETADR
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW	1
100	0033	LENGTH	RESW	1
105	0036	BUFFER	RESB	4096
106	1036	BUFFEND	EQU	*
107	1000	MAXLEN	EQU	BUFEND-BUFFER
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125	1036	RDREC	CLEAR	X
130	1038		CLEAR	A
132	103A		CLEAR	S
133	103C		+LDT	#MAXLEN
135	1040	RLOOP	TD	INPUT
140	1043		JEQ	RLOOP
145	1046		RD	INPUT
150	1049		COMPR	A,S
155	104B		JEQ	EXIT
160	104E		STCH	BUFFER,X
165	1051		TIXR	T
170	1053		JLT	RLOOP
175	1056	EXIT	STX	LENGTH
180	1059		RSUB	
185	105C	INPUT	BYTE	X'F1'
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210	105D	WRREC	CLEAR	X
212	105F		LDT	LENGTH
215	1062	WLOOP	TD	=X'05'
220	1065		JEQ	WLOOP
225	1068		LDCH	BUFFER,X
230	106B		WD	=X'05'
235	106E		TIXR	T
240	1070		JLT	WLOOP
245	1073		RSUB	
255	-----		END	FIRST
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
13	0003		LDB	#LENGTH
14	-----		BASE	LENGTH
15	0006	CLOOP	+JSUB	RDREC
20	000A		LDA	LENGTH
25	000D		COMP	#0
30	0010		JEQ	ENDFIL
35	0013		+JSUB	WRREC
40	0017		J	CLOOP
45	001A	ENDFIL	LDA	=C'EOF'
50	001D		STA	BUFFER
55	0020		LDA	#3
60	0023		STA	LENGTH
65	0026		+JSUB	WRREC
70	002A		J	@RETADR
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW	1
100	0033	LENGTH	RESW	1
105	0036	BUFFER	RESB	4096
106	1036	BUFFEND	EQU	*
107	1000	MAXLEN	EQU	BUFEND-BUFFER
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125	1036	RDREC	CLEAR	X
130	1038		CLEAR	A
132	103A		CLEAR	S
133	103C		+LDT	#MAXLEN
135	1040	RLOOP	TD	INPUT
140	1043		JEQ	RLOOP
145	1046		RD	INPUT
150	1049		COMPR	A,S
155	104B		JEQ	EXIT
160	104E		STCH	BUFFER,X
165	1051		TIXR	T
170	1053		JLT	RLOOP
175	1056	EXIT	STX	LENGTH
180	1059		RSUB	
185	105C	INPUT	BYTE	X'F1'
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210	105D	WRREC	CLEAR	X
212	105F		LDT	LENGTH
215	1062	WLOOP	TD	=X'05'
220	1065		JEQ	WLOOP
225	1068		LDCH	BUFFER,X
230	106B		WD	=X'05'
235	106E		TIXR	T
240	1070		JLT	WLOOP
245	1073		RSUB	
255	-----		END	FIRST
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	-----
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	032010
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	-----
	002D	*	=C'EOF'	454F46
95	0030	RETADR	RESW 1	-----
100	0033	LENGTH	RESW 1	-----
105	0036	BUFFER	RESB 4096	-----
106	1036	BUFFEND	EQU *	??
107	1000	MAXLEN	EQU BUFEND-BUFFER	
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
13	0003		LDB	#LENGTH
14	-----		BASE	LENGTH
15	0006	CLOOP	+JSUB	RDREC
20	000A		LDA	LENGTH
25	000D		COMP	#0
30	0010		JEQ	ENDFIL
35	0013		+JSUB	WRREC
40	0017		J	CLOOP
45	001A	ENDFIL	LDA	=C'EOF'
50	001D		STA	BUFFER
55	0020		LDA	#3
60	0023		STA	LENGTH
65	0026		+JSUB	WRREC
70	002A		J	@RETADR
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW	1
100	0033	LENGTH	RESW	1
105	0036	BUFFER	RESB	4096
106	1036	BUFFEND	EQU	*
107	1000	MAXLEN	EQU	BUFEND-BUFFER
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR	X
130	1038		CLEAR	A
132	103A		CLEAR	S
133	103C		+LDT	#MAXLEN
135	1040	RLOOP	TD	INPUT
140	1043		JEQ	RLOOP
145	1046		RD	INPUT
150	1049		COMPR	A,S
155	104B		JEQ	EXIT
160	104E		STCH	BUFFER,X
165	1051		TIXR	T
170	1053		JLT	RLOOP
175	1056	EXIT	STX	LENGTH
180	1059		RSUB	
185	105C	INPUT	BYTE	X'F1'
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR	X
212	105F		LDT	LENGTH
215	1062	WLOOP	TD	=X'05'
220	1065		JEQ	WLOOP
225	1068		LDCH	BUFFER,X
230	106B		WD	=X'05'
235	106E		TIXR	T
240	1070		JLT	WLOOP
245	1073		RSUB	
255	-----		END	FIRST
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
13	0003		LDB	#LENGTH
14	-----		BASE	LENGTH
15	0006	CLOOP	+JSUB	RDREC
20	000A		LDA	LENGTH
25	000D		COMP	#0
30	0010		JEQ	ENDFIL
35	0013		+JSUB	WRREC
40	0017		J	CLOOP
45	001A	ENDFIL	LDA	=C'EOF'
50	001D		STA	BUFFER
55	0020		LDA	#3
60	0023		STA	LENGTH
65	0026		+JSUB	WRREC
70	002A		J	@RETADR
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW	1
100	0033	LENGTH	RESW	1
105	0036	BUFFER	RESB	4096
106	1036	BUFFEND	EQU	*
107	1000	MAXLEN	EQU	BUFEND-BUFFER
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR	X
130	1038		CLEAR	A
132	103A		CLEAR	S
133	103C		+LDT	#MAXLEN
135	1040	RLOOP	TD	INPUT
140	1043		JEQ	RLOOP
145	1046		RD	INPUT
150	1049		COMPR	A,S
155	104B		JEQ	EXIT
160	104E		STCH	BUFFER,X
165	1051		TIXR	T
170	1053		JLT	RLOOP
175	1056	EXIT	STX	LENGTH
180	1059		RSUB	
185	105C	INPUT	BYTE	X'F1'
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR	X
212	105F		LDT	LENGTH
215	1062	WLOOP	TD	=X'05'
220	1065		JEQ	WLOOP
225	1068		LDCH	BUFFER,X
230	106B		WD	=X'05'
235	106E		TIXR	T
240	1070		JLT	WLOOP
245	1073		RSUB	
255	-----		END	FIRST
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	-----
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	032010
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	-----
	002D	*	=C'EOF'	454F46
95	0030	RETADR	RESW 1	-----
100	0033	LENGTH	RESW 1	-----
105	0036	BUFFER	RESB 4096	-----
106	1036	BUFFEND	EQU *	-----
107	1000	MAXLEN	EQU BUFEND-BUFFER	-----
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	F1
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	??
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	-----
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	-----
	1076	*	=X'05'	-----

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Pass Two

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	-----
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	032010
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	-----
	002D	*	=C'EOF'	454F46
95	0030	RETADR	RESW 1	-----
100	0033	LENGTH	RESW 1	-----
105	0036	BUFFER	RESB 4096	-----
106	1036	BUFEND	EQU *	-----
107	1000	MAXLEN	EQU BUFEND-BUFFER	-----
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	F1
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	274000
215	1062	WLOOP	TD X'05'	E32011
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	??
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	
	1076	*	=X'05'	

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

$$1076 - 1065 = 011$$

Address of literal - PC

Opcode 6 bits

n i x b p e displacement

1 1 1 0 0 0 1 1 0 0 1 0 (011)

Line	Loc	Source statement		Object code
5	0000	COPY	START 0	
10	0000	FIRST	STL RETADR	17202D
13	0003		LDB #LENGTH	69202D
14	-----		BASE LENGTH	-----
15	0006	CLOOP	+JSUB RDREC	4B101036
20	000A		LDA LENGTH	032026
25	000D		COMP #0	290000
30	0010		JEQ ENDFIL	332007
35	0013		+JSUB WRREC	4B10105D
40	0017		J CLOOP	3F2FEC
45	001A	ENDFIL	LDA =C'EOF'	032010
50	001D		STA BUFFER	0F2016
55	0020		LDA #3	010003
60	0023		STA LENGTH	0F200D
65	0026		+JSUB WRREC	4B10105D
70	002A		J @RETADR	3E2003
93	-----		LTORG	-----
	002D	*	=C'EOF'	454F46
95	0030	RETADR	RESW 1	-----
100	0033	LENGTH	RESW 1	-----
105	0036	BUFFER	RESB 4096	-----
106	1036	BUFEND	EQU *	-----
107	1000	MAXLEN	EQU BUFEND-BUFFER	-----
110		.		
115		.	SUBROUTINE TO READ RECORD INTO BUFFER	
120		.		
125	1036	RDREC	CLEAR X	B410
130	1038		CLEAR A	B400
132	103A		CLEAR S	B440
133	103C		+LDT #MAXLEN	75101000
135	1040	RLOOP	TD INPUT	E32019
140	1043		JEQ RLOOP	332FFA
145	1046		RD INPUT	DB2013
150	1049		COMPR A,S	A004
155	104B		JEQ EXIT	332008
160	104E		STCH BUFFER,X	57C003
165	1051		TIXR T	B850
170	1053		JLT RLOOP	3B2FEA
175	1056	EXIT	STX LENGTH	134000
180	1059		RSUB	4F0000
185	105C	INPUT	BYTE X'F1'	F1
195		.		
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER	
205		.		
210	105D	WRREC	CLEAR X	B410
212	105F		LDT LENGTH	774000
215	1062	WLOOP	TD =X'05'	E32011
220	1065		JEQ WLOOP	332FFA
225	1068		LDCH BUFFER,X	53C003
230	106B		WD =X'05'	DF2008
235	106E		TIXR T	B850
240	1070		JLT WLOOP	3B2FEF
245	1073		RSUB	4F0000
255	-----		END FIRST	??
	1076	*	=X'05'	

Pass Two

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

1076-106E =008

Address of literal – PC

Opcode 6 bits						n	i	x	b	p	e	displacement
1	1	0	1	1	1	1	1	0	0	1	0	(008)

Line	Loc	Source statement		Object code
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
13	0003		LDB	#LENGTH
14	-----		BASE	LENGTH
15	0006	CLOOP	+JSUB	RDREC
20	000A		LDA	LENGTH
25	000D		COMP	#0
30	0010		JEQ	ENDFIL
35	0013		+JSUB	WRREC
40	0017		J	CLOOP
45	001A	ENDFIL	LDA	=C'EOF'
50	001D		STA	BUFFER
55	0020		LDA	#3
60	0023		STA	LENGTH
65	0026		+JSUB	WRREC
70	002A		J	@RETADR
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW	1
100	0033	LENGTH	RESW	1
105	0036	BUFFER	RESB	4096
106	1036	BUFFEND	EQU	*
107	1000	MAXLEN	EQU	BUFEND-BUFFER
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR	X
130	1038		CLEAR	A
132	103A		CLEAR	S
133	103C		+LDT	#MAXLEN
135	1040	RLOOP	TD	INPUT
140	1043		JEQ	RLOOP
145	1046		RD	INPUT
150	1049		COMPR	A,S
155	104B		JEQ	EXIT
160	104E		STCH	BUFFER,X
165	1051		TIXR	T
170	1053		JLT	RLOOP
175	1056	EXIT	STX	LENGTH
180	1059		RSUB	
185	105C	INPUT	BYTE	X'F1'
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR	X
212	105F		LDT	LENGTH
215	1062	WLOOP	TD	=X'05'
220	1065		JEQ	WLOOP
225	1068		LDCH	BUFFER,X
230	106B		WD	=X'05'
235	106E		TIXR	T
240	1070		JLT	WLOOP
245	1073		RSUB	
255	-----		END	FIRST
	1076	*	=X'05'	??

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076

Line	Loc	Source statement		Object code
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
13	0003		LDB	#LENGTH
14	-----		BASE	LENGTH
15	0006	CLOOP	+JSUB	RDREC
20	000A		LDA	LENGTH
25	000D		COMP	#0
30	0010		JEQ	ENDFIL
35	0013		+JSUB	WRREC
40	0017		J	CLOOP
45	001A	ENDFIL	LDA	=C'EOF'
50	001D		STA	BUFFER
55	0020		LDA	#3
60	0023		STA	LENGTH
65	0026		+JSUB	WRREC
70	002A		J	@RETADR
93	-----		LTORG	
	002D	*	=C'EOF'	
95	0030	RETADR	RESW	1
100	0033	LENGTH	RESW	1
105	0036	BUFFER	RESB	4096
106	1036	BUFFEND	EQU	*
107	1000	MAXLEN	EQU	BUFEND-BUFFER
110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	1036	RDREC	CLEAR	X
130	1038		CLEAR	A
132	103A		CLEAR	S
133	103C		+LDT	#MAXLEN
135	1040	RLOOP	TD	INPUT
140	1043		JEQ	RLOOP
145	1046		RD	INPUT
150	1049		COMPR	A,S
155	104B		JEQ	EXIT
160	104E		STCH	BUFFER,X
165	1051		TIXR	T
170	1053		JLT	RLOOP
175	1056	EXIT	STX	LENGTH
180	1059		RSUB	
185	105C	INPUT	BYTE	X'F1'
195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	105D	WRREC	CLEAR	X
212	105F		LDT	LENGTH
215	1062	WLOOP	TD	=X'05'
220	1065		JEQ	WLOOP
225	1068		LDCH	BUFFER,X
230	106B		WD	=X'05'
235	106E		TIXR	T
240	1070		JLT	WLOOP
245	1073		RSUB	
255	-----		END	FIRST
	1076	*	=X'05'	

Pass Two

Symbol Table

Symbol	Type	Value
FIRST	R	0000
CLOOP	R	0006
ENDFIL	R	001A
RETADR	R	0030
LENGTH	R	0033
BUFFER	R	1036
BUFFEND	R	1036
MAXLEN	A	1000
RDREC	R	1036
RLOOP	R	1040
EXIT	R	1056
INPUT	R	105C
WRREC	R	105D
WLOOP	R	1062

Literal Table

Name	value	Address
=C'EOF'	454F46	002D
=X'05'	05	1076